Physics-Gated Correction Discovery: Why Constraining the Search Space Beats Scaling It in Noisy Symbolic Regression

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Abstract

Symbolic regression (SR) methods such as PySR, AI Feynman, and PhySO discover equations *from scratch*, an approach that is sensitive to noisy physical data: under matched residuals, PySR's structural accuracy is non-monotonic in noise and does not improve when its search budget is doubled. We argue that scientific discovery is rarely *tabula rasa*—it is *correction-first*: given a known classical law and anomalous data, the real task is to find the minimal physically-valid correction Δ. We present **Anomaly-Driven Correction Discovery (ADCD)**, which restricts search to a correction subspace and cascades three physical filters—AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency (ARC)—*before* a JAX optimizer fits any parameter, with Bayesian Information Criterion (BIC) reranking. On a 9-scenario benchmark (overall mean $80.4\% \pm 7.4\%$ across 16 seeds and all noise levels; 88.9% at the disclosed reference seed), ADCD reaches 88.9% at 5% noise versus 11.1% for PySR—a *seed-independent* gap: ADCD's *worst* of 16 seeds still exceeds PySR with doubled budget. Removing all gates drops recovery by 44.5 pp. ADCD is orthogonal to—not competitive with—existing SR: it restricts the problem they solve to a physically-motivated subspace where unconstrained search is least reliable under noise.

1 Introduction

When a physical law breaks down under extreme conditions, the scientific question is not "what is the correct law?"—it is "what is the *minimal correction* to the known law?" The dominant paradigm in symbolic regression (SR) answers a different question: it discovers $f(X) \approx y$ from a blank slate, an approach that—as we show—degrades under realistic noise. Pioneered by Eureqa Schmidt & Lipson [2009], AI Feynman Udrescu & Tegmark [2020], and PySR Cranmer [2020], and extended by deep RL approaches such as DSR Petersen et al. [2021] and unit-constrained PhySO Tenachi et al. [2023], these systems recover compact laws in *low*-noise regimes. Under realistic observational noise, however, their structural accuracy degrades and is often *non-monotonic* in noise: a larger candidate pool does not reliably translate into the correct functional *class* (Section 4).

We observe that this difficulty is partly one of *problem formulation*. Science rarely discovers from a blank slate—it *corrects*. From the relativistic kinetic-energy expansion $\frac{1}{2}mv^2(1 + \frac{3}{4}v^2/c^2 + \dots)$, to Debye-screened Coulomb potentials $V(r) \propto r^{-1}e^{-r/\lambda_D}$, to turbulent drag $F = bv + \alpha v^2$, progress takes a structurally-correct classical law and appends a *minimal correction* Δ that reconciles it with anomalous data. This restricts the hypothesis space from the full function space $\mathcal{F}$ to a correction subspace $\mathcal{D} \subset \mathcal{F}$—an exponential reduction informed by the known physics.

We propose **Anomaly-Driven Correction Discovery (ADCD)**, built on three contributions:

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C1 Correction-first paradigm. Search for Δ in $\mathcal{D}$, not f in $\mathcal{F}$, given a classical law $y_{\text{classical}} = g(X;\phi)$ and anomalous observations.

C2 Physics-gated cascade before optimization. Three filters—AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency (ARC, $\Delta \to 0$ at the classical boundary)—screen candidates *before* any parameter is fit.

C3 Stability where unconstrained search is not. Because the correction-first regime never enters the universal-approximator regime that produces nested $\sin(\sin(\dots))$ at 0% noise, ADCD's structural accuracy is monotonic in noise where tabula rasa search is non-monotonic.

We deliberately scope this paper to *method and synthetic benchmark*; real-data validation (galactic, cosmological) is reported in a companion manuscript.

2 Related work

Tabula rasa symbolic regression. Modern SR discovers an expression $f(X) \approx y$ from a blank slate. Eureqa Schmidt & Lipson [2009] pioneered genetic search over expression trees; AI Feynman Udrescu & Tegmark [2020] added physics-inspired unit filtering and progressive complexity; PySR Cranmer [2020] scaled genetic SR with Pareto-front selection; DSR Petersen et al. [2021] replaced the search with risk-seeking reinforcement learning. All share a common goal—recovering the *full* law—and a common limitation: under realistic noise their structural accuracy degrades and is often *non-monotonic* in noise, because a larger candidate pool rewards overfit structure rather than the correct functional class. ADCD departs by searching only the correction subspace $\mathcal{D} \subset \mathcal{F}$ to a *known* law, an exponential restriction of the hypothesis space informed by prior physics.

Unit-constrained SR. PhySO Tenachi et al. [2023] enforces strict dimensional homogeneity throughout training, recovering full equations from data with strong results in clean regimes. Its constraints operate *during* optimization; ADCD's physics gates screen candidates *before* any fit, and it relaxes parameter units adaptively rather than enforcing them strictly. The two are complementary design points: ADCD could take a PhySO-style proposer and feed it through its gate cascade, combining strict-unit generation with pre-fit asymptotic and complexity filtering.

LLM- and concept-library-driven SR. LaSR Grayeli et al. [2024] uses a learned concept library, prompted by a large language model, to improve the *generation* of candidate expressions, but applies no physics filtering before fitting. ADCD is orthogonal: its gate cascade can act as a selection layer on top of any proposer—genetic, RL, or LLM-driven—subjecting generated templates to the same dimensional, asymptotic, and complexity constraints regardless of how they were proposed.

Physics embedded in the model vs. the search space. A separate line of work embeds physical structure in the *model or training procedure* itself: physics-informed neural networks Raissi et al. [2019] encode governing PDEs into the loss, Hamiltonian neural networks Greydanus et al. [2019] constrain dynamics to a conserved Hamiltonian, and SINDy Champion et al. [2019] sparse-regresses dynamical-system coefficients. ADCD occupies an orthogonal axis: rather than baking physics into a learned model, it restricts the symbolic *search space* via interpretable, pre-fit gates that any downstream optimizer—JAX or otherwise—then operates within.

3 The ADCD framework

Residual extraction. Given a known classical law $y_{\text{classical}} = g(X;\phi)$ and noisy anomalous data y_{obs}, ADCD isolates the correction by forming a residual. In *multiplicative* mode $y_{\text{true}} = y_{\text{classical}}(1 + \Delta)$ gives $\delta = y_{\text{obs}}/y_{\text{classical}} - 1$; in *additive* mode $y_{\text{true}} = y_{\text{classical}} + \Delta$ gives $\delta = y_{\text{obs}} - y_{\text{classical}}$. A mode selector picks whichever best isolates the anomalous deviation.

Proposer. Candidate correction templates $\Delta(X;\theta)$ are drawn from a stochastic sampler over dimensionless ratios, biased by residual features (monotonicity, curvature, oscillation, exponential decay R^2, symmetry) that upweight plausible families but never gate candidates—only the physics cascade does.

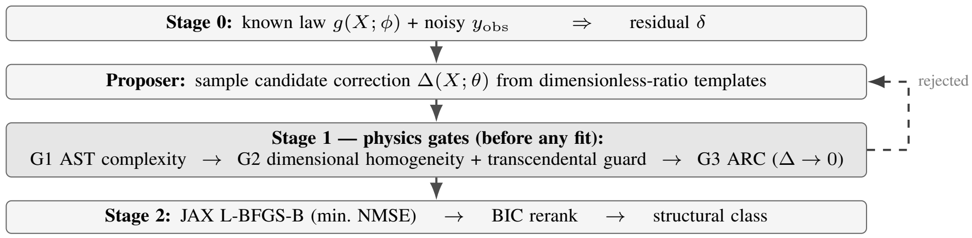


Figure 1: ADCD pipeline. Physics gates (G1–G3) screen candidates *before* any parameter is fit; rejected candidates are re-sampled.

Stage 1: cascaded physics gates. Every candidate passes three filters *before* optimization:

G1. AST complexity. Reject expressions with AST depth > 7 or > 20 tokens—a coarse Occam prior.

G2. Dimensional homogeneity + transcendental guard. Verify Δ is dimensionless; physical variables may enter only inside dimensionless ratios. As a sub-check, $\sin, \cos, \exp, \log, \tanh$ may take only *dimensionless* arguments: $\exp(-r)$ without a scale θ_1 is blocked instantly. Free parameters carry adaptive units (relaxable), documented in the companion paper.

G3. Asymptotic-regime consistency (ARC). Enforce the decoupling limit: as $x \to x_0$ (e.g. $v \to 0$, $r \to \infty$), $\lim_{x\to x_0} \Delta(X;\theta) = 0$, evaluated with SymPy's limit engine.

Survivors are then coarsely ranked by preliminary NMSE on δ, giving an early data-driven signal before the full fit.

Stage 2: JAX-traced optimization + BIC reranking. Surviving expressions are transpiled to JAX and fit by L-BFGS-B (multi-restart, log-uniform init) minimizing NMSE of the residual, $\text{NMSE} = \frac{\sum_i(\Delta(X_i;\theta)-\delta_i)^2}{\sum_i(\delta_i-\bar{\delta})^2}$. Under noise, flexible wrong-class expressions can achieve lower NMSE than the truth, so we rerank by the Bayesian Information Criterion Schwarz [1978], Rissanen [1978], $\text{BIC} = N\ln(\text{NMSE}) + k\ln N$, whose $k \ln N$ penalty selects parsimonious correct-class forms. The winner is classified into one of six families (exponential, trigonometric, logarithmic, power law, rational, polynomial) by AST traversal, with degenerate-exponent detection (a fitted $\hat{\theta}_1 \approx 1$ recasts $\theta_0 x^{\theta_1}$ as polynomial).

Relation to EFT. The cascade mirrors the correction-first philosophy of Effective Field Theory Weinberg [1979]: dimensional analysis of operators (G2), decoupling of heavy modes in the low-energy limit (G3/ARC), and power counting (BIC). ADCD has no renormalization-group flow or controlled E/Λ expansion—it *automates* that philosophy for data-driven problems.

4 Experiments

Benchmark. We evaluate on 9 physical-anomaly scenarios in three tiers: *textbook* (relativistic KE, Yukawa gravity, anharmonic spring), *cross-domain* (net radiation T^{-4}, nonlinear drag, screened Coulomb), and *synthetic novel* formulas. Each is run at $0/1/5/10\%$ observational noise. Success = correct structural *class* recovery (e.g. discovering an exponential correction where the truth is exponential). We report a 16-seed distribution; the mean is the primary claim and seed=42 (the highest seed) is disclosed explicitly, never as the headline.

Baselines. We compare against PySR Cranmer [2020] on the *same residual targets* $\delta = y_{\text{obs}} - y_{\text{classical}}$. The primary baseline is the `fair` profile (n_{iter}=100, `maxsize`=30, 60 s/scenario). To rule out a budget artefact we add a `generous` profile that *doubles* every budget axis (n_{iter}=200, `maxsize`=40, 120 s; $\sim 3\times$ wall-clock).

Main result: the gap is structural, not budgetary. Table 1 reports structural match rates. ADCD reaches 88.9% (8/9) at 5% noise versus 11.1% (1/9) for PySR fair—a 77.8 pp gap under matched residuals. The comparison is *seed-independent*: at 5% noise ADCD's worst of 16 seeds (66.7%, 6/9; cf. the per-seed distribution in our bundled artefacts) still exceeds PySR fair. Crucially, doubling PySR's budget does *not* close the gap—`generous` total recovery stays flat at $15/36$ (identical to

Table 1: Structural class match rate on the 9-scenario residual benchmark. `generous` doubles PySR's budget on every axis yet does not close the gap.

Method	**0%**	**1%**	**5%**	**10%**
ADCD (ours, seed=42)	9/9 (100%)	9/9 (100%)	**8/9 (88.9%)**	8/9 (88.9%)
ADCD 16-seed mean	86.8%	81.3%	*77.1%±10.0%*	76.4%
PySR fair	4/9 (44.4%)	5/9 (55.6%)	1/9 (11.1%)	5/9 (55.6%)
PySR generous (2×)	4/9 (44.4%)	4/9 (44.4%)	5/9 (55.6%)	2/9 (22.2%)

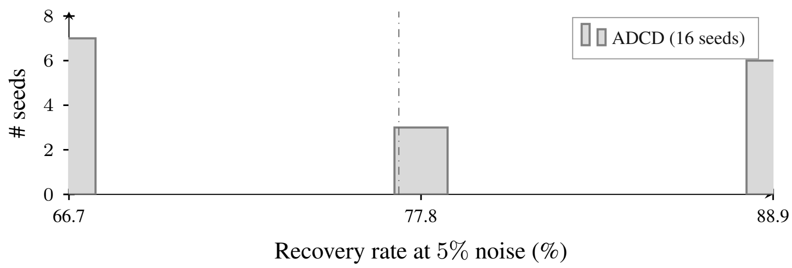


Figure 2: Seed-independence of the ADCD–PySR gap at 5% noise. Every one of ADCD's 16 seeds (bars) recovers at least 66.7%, above PySR's best result even under doubled budget (55.6%, dotted).

`fair`) and at the 5% headline point it reaches only $5/9$ (55.6%). The extra budget explores deeper trees (mean complexity 9.8 vs. 8.7 nodes) spent on more deeply-overfit rather than more physically-correct structure. The binding constraint is *structural selection*, not search budget; physics gates resolve it directly.

The seed-independence of this gap is structural, not artefactual. Figure 2 bins ADCD's per-seed recovery at the 5% headline noise level: the entire 16-seed distribution sits between 66.7% and 88.9%, *entirely above* PySR's best result even under doubled budget (55.6%, `generous`). ADCD's worst seed exceeds PySR's best by 11.1 pp without exception—the gap cannot be attributed to a favourable seed choice.

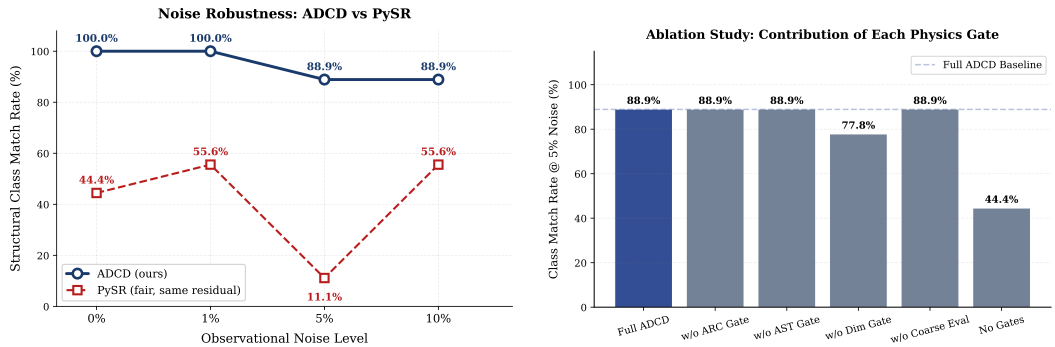
Why PySR is non-monotonic, and why ADCD is not. PySR fair exhibits a non-monotonic noise profile $(4, 5, 1, 5$ out of $9)$. At 0% noise the large search space produces deeply-nested trigonometric expressions (mean 13.8 nodes) that fit clean data but are structurally wrong ($5/5$ mismatches are trigonometric); at 1% noise regularization drops complexity to 8.2 nodes; at 5% it picks the wrong family entirely; at 10% it returns to the simplest form. ADCD avoids this because its gates are noise-independent: dimensional and ARC constraints reject the nested trig that PySR overfits to, and once the correct family is sampled the cascade selects it regardless of SNR (Figure 3a).

Ablation: the gates act collectively. Figure 3b shows the result of disabling gates one at a time at 5% noise. Removing ARC, AST, or the coarse-eval pre-filter individually leaves accuracy unchanged $(8/9)$; removing the dimensional gate alone costs 11.1 pp $(8/9 \to 7/9)$. The dimensional gate is the most critical single filter because violations of dimensional homogeneity—e.g. $\exp(-r)$ with r in metres—are the most common failure mode in unconstrained SR, while ARC violations are rarer but produce unphysical extrapolation. Only disabling *all* gates causes a larger collapse $(8/9 \to 4/9$, -44.5 pp): the gates reinforce each other as a collective physical-consistency filter rather than each being independently essential.

5 Discussion and limitations

ADCD is a restriction, not a competitor. ADCD does not compete with PhySO or PySR—it restricts the problem they solve to a physically-motivated subspace where unconstrained search is least reliable under noise. This makes ADCD directly *combinable* with them: a PhySO/PySR/LaSR Grayeli et al. [2024] proposer could feed the same gate cascade.

Honest limitations. ADCD assumes a structurally-correct classical baseline (incremental correction, not wholesale replacement). At $\geq 5\%$ noise, screened-Coulomb decay is fitted by a rational proxy,



(a) Noise robustness. ADCD stable; PySR fair non-monotonic (44→56→11→56%).

(b) Gate ablation at 5% noise. Removing all gates collapses recovery by 44.5 pp.

Figure 3: Left: structural match rate vs. noise. Right: per-gate ablation.

reflecting decay-vs-saturation ambiguity under limited dynamic range. A multivariable extension (Buckingham-Π) is currently weaker ($2/4$ at 5%) and treated as exploratory; the dimensional gate uses adaptive parameter-unit relaxation rather than PhySO's strict enforcement—complementary design points.

Reproducibility. ADCD ships as an open-source Python package (JAX backend, public API/CLI, CI/CD); all 16 seeds, per-seed noise distributions, and PySR profiles are bundled for replication.

Fail-safe behavior. On a 6-case misspecification benchmark, ADCD fails safe—it fails to converge rather than producing confident spurious corrections—a property absent from tabula rasa SR.

AI disclosure. Code implementation and manuscript preparation were assisted by large language models under a human-in-the-loop workflow. All scientific hypotheses, experimental designs, analysis, and conclusions were conceived and verified solely by the human author, who remains fully responsible for the work.

Closing. Noisy symbolic regression is bottlenecked by *structural selection*, not search budget. The correction-first formulation—searching for a physically-valid Δ to a known law, screened by dimensional and asymptotic gates before any fit—resolves this directly: ADCD delivers seed-independent gains over PySR, stays robust where unconstrained search is non-monotonic, and is orthogonal to existing SR systems. Future work extends the cascade to multivariable laws and integrates it as a selection layer over external proposers (PhySO, LaSR).

References

M. Schmidt & H. Lipson. Distilling free-form natural laws from experimental data. *Science*, 324(5923):81–85, 2009.

S. M. Udrescu & M. Tegmark. AI Feynman: A physics-inspired systematic approach to symbolic regression. *Science Advances*, 6(16), 2020.

M. Cranmer. PySR: Fast & parallelized symbolic regression in Python/Julia. *arXiv:2007.03738*, 2020.

B. K. Petersen et al. Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients. In *ICLR*, 2021.

W. Tenachi, R. Ibata, & F. I. Diakogiannis. Deep symbolic regression for physics guided by units constraints. *Astrophysical Journal*, 959(2):99, 2023.

A. Grayeli, A. Sehgal, O. Costilla-Reyes, M. Cranmer, & S. Chauduri. Symbolic regression with a learned concept library. In *NeurIPS*, 2024.

G. Schwarz. Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464, 1978.

J. Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978.

S. Weinberg. Phenomenological Lagrangians. *Physica A*, 96(1-2):327–340, 1979.

M. Raissi, P. Perdikaris, & G. E. Karniadakis. Physics-informed neural networks: A deep-learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

S. Greydanus, M. Dzamba, & J. Yosinski. Hamiltonian neural networks. In *NeurIPS*, 2019.

K. Champion, B. Lusch, J. N. Kutz, & S. L. Brunton. Data-driven discovery of coordinates and governing equations. *Proceedings of the National Academy of Sciences*, 116(45):22445–22451, 2019.